

MARS SOCIETY 2026

MOXIE Demo & Ground-Truth Package

A worked example of observed facts, bounded inference, and blocked overclaim.

THE INTELLIGENCE LAYER FOR MARS

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Verified demonstration

Claim	Class	Status
16 oxygen-producing runs	Observed	Verified to NASA mission report
122 g total oxygen	Observed	Verified to NASA mission report
Peak 12 g/h at $\geq 98\%$ purity	Observed	Verified to NASA mission report
Crew-scale oxygen plant is flight-ready	Unknown	Blocked; not demonstrated

Ground-truth gate

The gate checks source identity, claim wording, units, freshness, reproducibility, independence, and whether conclusions exceed the evidence. It returns conditional pass: MOXIE validated oxygen extraction from the Martian atmosphere across seasons, but system scale-up, dust tolerance, continuous duty, power integration, compression or liquefaction, storage, maintainability, and qualification remain future engineering work.

Promotion packet

The bundled JSON stage packet preserves the verified claims, blocks the readiness overclaim, assigns the next action to an ISRU program lead, and requests a scale-up requirements and integrated storage/liquefaction trade study.

Authoritative source

NASA, "NASA's Oxygen-Generating Experiment MOXIE Completes Mars Mission," 6 September 2023: <https://www.nasa.gov/missions/mars-2020-perseverance/perseverance-rover/nasas-oxygen-generating-experiment-moxie-completes-mars-mission/>